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International Conference on Intelligent Computing and Sustainable Technologies : Submission (199) has been edited.

Microsoft CMT <noreply@msr-cmt.org> 2 May 2026 at 14:19
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Hello,

The following submission has been edited.

Track Name: IntCST2026

Paper ID: 199

Paper Title: Automated Breast Cancer Detection Using Attention-Based CNN

Abstract:
Breast cancer remains a primary cause of cancer-related deaths for women worldwide, highlighting the critical need for timely and accurate diagnosis to enhance survival rates. Traditional diagnostic methods that depend on interpreting clinical images like mammograms and ultrasounds are increasingly hindered by variations in tumor characteristics such as shape, size, and density, as well as imaging noise, which lead to inconsistent diagnoses. This study introduces an advanced deep learning framework for automated breast cancer detection that combines an attention-based CNN with transfer learning. The methodology incorporates preprocessing techniques such as normalization, resizing, contrast enhancement, and data augmentation to improve model robustness. A pre-trained CNN is fine-tuned to extract high-level features, while an attention mechanism focuses on tumor-relevant regions and suppresses background noise. The extracted features undergo processing through globally standard pooling and fully connected layers for binary classification. The model is assessed using accuracy, precision, recall, F1-score, and AUC. Experimental results show improved overall performance compared to traditional CNN models, with higher sensitivity and specificity. The suggested framework provides a dependable laptop-based diagnostic tool that enhances early detection and aids medical decision-making.

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Submission Questions Response:
1. Conflict of interest
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6. Authors Contributions

Harshit, Aakriti Shory, Bhoomika, Isha Kansal; methodology development: Harshit, Aakriti Shory, Bhoomika, Isha Kansal; data collection and dataset preparation: Harshit, Bhoomika, Isha Kansal; implementation and model development: Harshit, Aakriti Shory, Isha Kansal; analysis and interpretation of results: Harshit, Bhoomika, Aakriti Shory, Isha Kansal; draft manuscript preparation: Harshit, Aakriti Shory, Isha Kansal; review and editing: Harshit, Bhoomika, Isha Kansal; supervision and coordination: Harshit, Isha Kansal. All authors reviewed the results and approved the final version of the manuscript.

7. Conference Rules and Guidelines

Agreement accepted

Thanks,
CMT team.